

PSS Week 5 P12

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1 Proof

Solved with An Vu. Details filled in by me, I suppose.

Lemma: Suppose we have n starting and ending cells and end with one fewer inmate. Then some inmate ends up having fewer cellmates.

Proof: Suppose for the sake of contradiction that each inmate has at least as many cellmates after the change. Let a_i denote the number of starting cells containing exactly i inmates. Denote b_i similarly for the ending cells. Then considering the number of inmates before and after,

$$\sum_{i=1}^n i \cdot a_i = \sum_{i=1}^n i \cdot b_i + 1 \quad (1).$$

Considering the ending and starting cells:

$$\sum_{i=0}^n a_i = \sum_{i=1}^n b_i. \quad (2)$$

Note the left sum starts from $i = 0$, while the right one starts from $i = 1$ as the ending cells cannot be empty. Considering the number of inmates with at most $j - 1$ other cellmates for arbitrary j ,

$$\sum_{i=1}^j i \cdot a_i \geq \sum_{i=1}^j i \cdot b_i. \quad (3)$$

We may now manipulate these results to get a contradiction. Multiply equation (3) by $\frac{1}{j(j+1)}$ and sum over j :

$$\begin{aligned} \sum_{j=1}^n \frac{1}{j(j+1)} \sum_{i=1}^j i \cdot a_i &\geq \sum_{j=1}^n \frac{1}{j(j+1)} \sum_{i=1}^j i \cdot b_i \longrightarrow \\ \sum_{i=1}^n \sum_{j=i}^n \frac{i}{j(j+1)} a_i &\geq \sum_{i=1}^n \sum_{j=i}^n \frac{i}{j(j+1)} b_i \longrightarrow \end{aligned}$$

$$\sum_{i=1}^n i \left(\frac{1}{i} - \frac{1}{n} \right) a_i \geq \sum_{i=1}^n i \left(\frac{1}{i} - \frac{1}{n} \right) b_i \longrightarrow$$

$$\sum_{i=1}^n \left(1 - \frac{i}{n} \right) a_i \geq \sum_{i=1}^n \left(1 - \frac{i}{n} \right) b_i \quad (4).$$

We may now ignore a_0 in (2) to get $\sum b_i \geq \sum a_i$, so combining with (4) gives

$$\sum_{i=1}^n \frac{i}{n} b_i \geq \sum_{i=1}^n \frac{i}{n} a_i,$$

contradicting (1).

Now consider the (arbitrary) $n+1$ th cell. Treat an inmate going to that cell as if they disappear. Then apply our lemma to the n starting and ending cells. There exists an inmate with fewer cellmates, and finishes at cell k . Apply the lemma to the cells excluding cell k and we have our second inmate, and we're done.